Multi-Agent Reinforcement Learning

End Semester Examination
Find the codes here : MARL

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1 Question 1 — Multi-Agent Independent Q-Learning

(a) Scenario Parameters.

Parameter	Value	Derivation
Number of agents	2	Fixed
Grid world size	10×10	$5 + 5$ (last digit = 5)
Goal condition	Simultaneous arrival	Both agents must reach goal together
Reward	+10 if both reach, else 0	Sparse reward
Episode length	30 steps	Fixed
Training episodes	2185	Last 4 digits
Modification type	Reward Shaping	$\Gamma = 5 \Rightarrow$ otherwise case

The environment is a cooperative 10×10 grid world. Agent 1 starts at $(0, 0)$ and Agent 2 starts at $(0, 9)$, while the goal is at $(9, 9)$. A reward is obtained **only when both agents reach the goal simultaneously**, making the task a coordination problem rather than a simple navigation problem.

Background: Independent Q-Learning. Each agent learns its own Q-function independently:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \tag{1}$$

Each agent:

- Observes only its own state
- Treats the other agent as part of the environment
- Uses ε-greedy exploration

Exploration Policy.

$$\pi(a|s) = \begin{cases} \arg\max_a Q(s, a), & \text{with probability } 1 - \varepsilon \\ \text{random action}, & \text{with probability } \varepsilon \end{cases} \tag{2}$$

$$\varepsilon_t = \max(\varepsilon_{\min}, \varepsilon_0 \cdot \delta^t) \tag{3}$$

Parameter	Value	Description
α	0.1	Learning rate
γ	0.95	Discount factor
ε_0	1.0	Initial exploration
$\varepsilon_{\min}$	0.05	Minimum exploration
Decay rate	0.995	Exploration decay

Hyperparameters.

(b) Baseline Independent Q-Learning.

Results Summary.

- Final success rate: **0.00%**
- Reward across all episodes: **0**
- ε reaches minimum by $\sim$600 episodes

Learning Curve.

Key Observation (Non-Trivial Pattern).

The learning curve remains flat at zero reward throughout training despite successful exploration decay.

Explanation:

1. **Sparse Reward Problem:** The only reward is received when both agents reach the goal *simultaneously*. This event is extremely rare during random exploration.
2. **Premature Convergence:** ε decays quickly and reaches 0.05 by $\sim$600 episodes. Since no successful trajectory is found during high exploration, both agents converge to near-zero Q-values.
3. **Coordination Failure:** Even if agents independently learn paths to the goal, they fail to synchronize arrival times.
4. **Asymmetric Start Positions:** Agent 1 requires $\sim$ 18 steps, while Agent 2 requires $\sim$ 9 steps. This makes simultaneous arrival inherently difficult.

Conclusion: Independent Q-learning fails due to **sparse rewards + lack of coordination + premature exploitation**.

(c) Modification: Reward Shaping. Since $\Gamma = 5$, reward shaping is applied.

Potential-Based Reward Shaping.

$$F(s, s') = \gamma\Phi(s') - \Phi(s) \tag{4}$$

$$\tilde{r} = r + F(s, s') \tag{5}$$

Potential function:

$$\Phi(s) = -\left(|r - r_g| + |c - c_g|\right) \tag{6}$$

This represents the negative Manhattan distance to the goal.

Results Summary.

- Overall success rate: **88.24%**
- Rapid improvement after $\sim$200 episodes
- Near-optimal performance after $\sim$600 episodes

Learning Curve (With Reward Shaping).

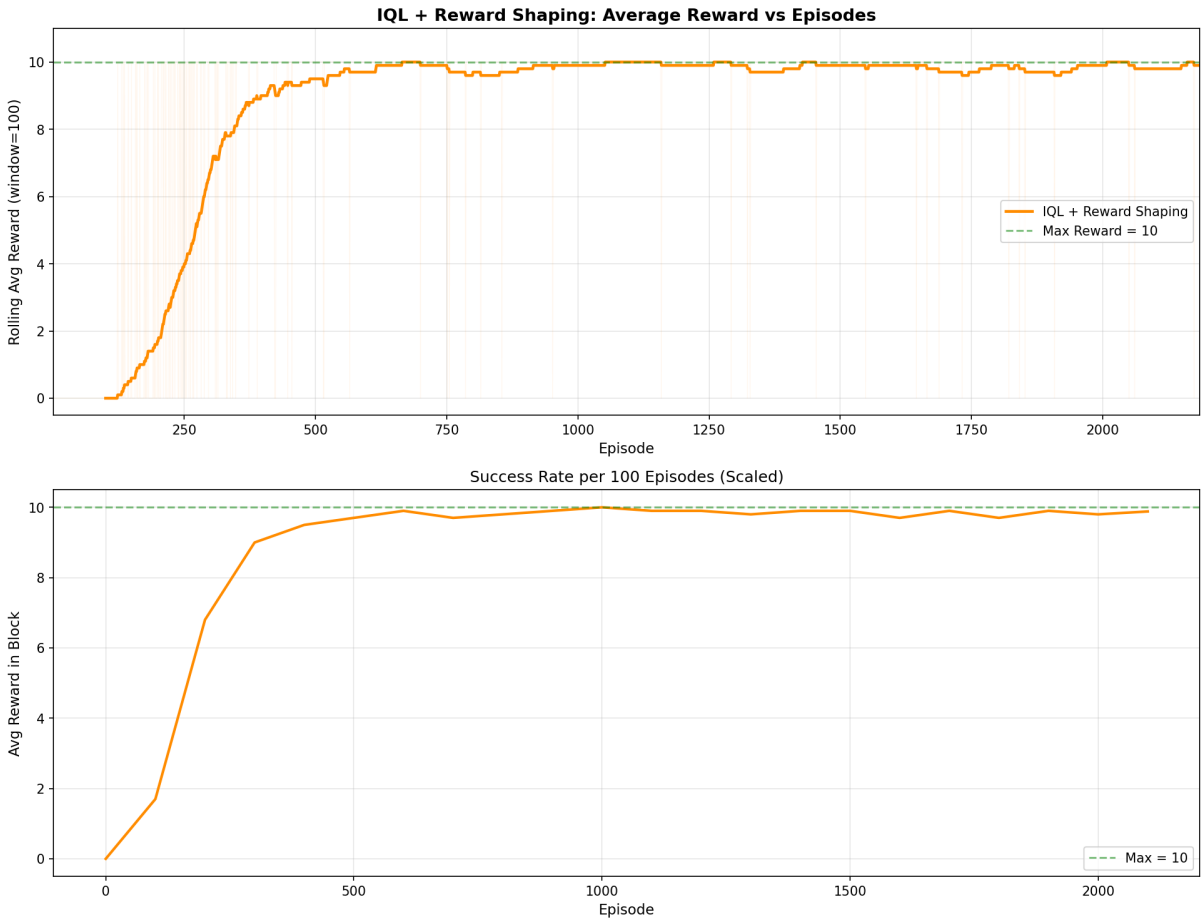

Figure 2: IQL with reward shaping: Rapid increase in average reward, showing successful learning and coordination.

Observed Learning Behavior.

- Episode 200: ∼8.5% success
- Episode 400: ∼79% success
- Episode 600+: ∼96–99% success

Aspect	Baseline	With Shaping
Reward signal	Sparse	Dense
Learning speed	Very slow	Fast
Coordination emergence	Fails	Successful
Q-value updates	Rare	Frequent

What Changed and Why.

1. **Dense Feedback:** Every step toward the goal provides a positive signal, even without reaching the goal.
2. **Directional Learning:** Agents learn which actions reduce distance to the goal early in training.
3. **Improved Exploration Efficiency:** Even random exploration produces meaningful updates.
4. **Faster Coordination:** Since both agents learn goal-directed behavior faster, the probability of simultaneous arrival increases.

5. **Policy Invariance:** The shaping function does not change the optimal policy, only accelerates learning.

Key Insight.

Reward shaping converts a sparse-reward coordination problem into a dense-learning problem, enabling both agents to learn useful policies early and eventually coordinate successfully.

Final Comparison.

- Baseline IQL: **No learning, 0% success**
- IQL + Reward Shaping: **Fast convergence, ∼88% success**

Baseline vs Reward Shaping. **Overall Conclusion:**
Independent Q-learning alone is insufficient for multi-agent coordination under sparse rewards. Reward shaping provides the necessary intermediate signal to guide learning, significantly improving both convergence speed and final performance.

2 Question 2 — Communication Constraint Problem

Problem Setup. This question uses the same cooperative 10×10 grid-world from Question 1, but now each agent is allowed to send **one binary message** $m \in \{0, 1\}$ at every time-step. The communication mechanism is added on top of the reward-shaped Independent Q-Learning setup from Q1 so that the effect of communication can be isolated cleanly.
For comparison:

- **No-communication baseline:** IQL + reward shaping
- **Modified setting:** IQL + reward shaping + 1-bit communication

The training setup remains the same as in Q1:

- Grid size: 10×10
- Episode length: 30 steps
- Training episodes: 2185
- Learning rate: $\alpha = 0.1$
- Discount factor: $\gamma = 0.95$
- Exploration: ε-greedy with decay

(a) Communication Strategy.

1-bit message interpretation: Goal-proximity signal

$m_i^t = 1$ means *"agent i is close to the goal"*
$m_i^t = 0$ means *"agent i is still far from the goal"*

Threshold used in the code: Manhattan distance ≤ 3

Timing: The bit is computed at every time-step and appended to the partner's state before action selection.

Formally, the message sent by agent i at time t is

$$m_i^t = \mathbf{1}\left[\|\text{pos}_i^t - \text{goal}\|_1 \leq 3\right]. \tag{7}$$

The augmented input to each agent is therefore

$$\tilde{s}_i^t = (x_i^t, y_i^t, m_j^t), \tag{8}$$

where m_j^t is the message received from the partner.

Why this strategy? The goal of communication here is not to transmit the exact position of the partner, but only a coarse progress indicator. In a simultaneous-arrival task, the most useful high-level question is often: *Is my partner almost done, or still far away?* This 1-bit summary is simple, cheap, and easy to integrate into tabular Q-learning.

(b) Implementation and Results.

Learning Setup. The agents are trained with the same Independent Q-Learning update as in Q1. The only difference is that each agent now conditions its Q-value update on the partner's communication bit.
The shaped reward used in both cases is

$$\tilde{r}_i^t = r^t + \gamma\Phi(s_i^{t+1}) - \Phi(s_i^t), \tag{9}$$

where the potential function is the negative Manhattan distance to the goal:

$$\Phi(s_i^t) = -\left(|x_i^t - x_g| + |y_i^t - y_g|\right). \tag{10}$$

So the comparison is:

- **No communication:** state $= (x_i, y_i)$
- **With communication:** state $= (x_i, y_i, m_j)$

Learning Curves.

Observed Results. The numerical results from the code are:

Metric	No Communication	1-bit Communication
Overall success rate (%)	88.24	88.74
Final 100-episode success rate (%)	99.00	97.00
Peak rolling average reward	10.000	10.000
Q-table entries for Agent 1	98	188

Interpretation of the plot.

- Both curves reach the maximum reward of 10, so both versions learn the task successfully.
- The communication curve is slightly better in the early and middle phases, which is visible in the top and bottom panels.
- The improvement is small, and the final-100-episode success rate is actually slightly lower for the communication version.

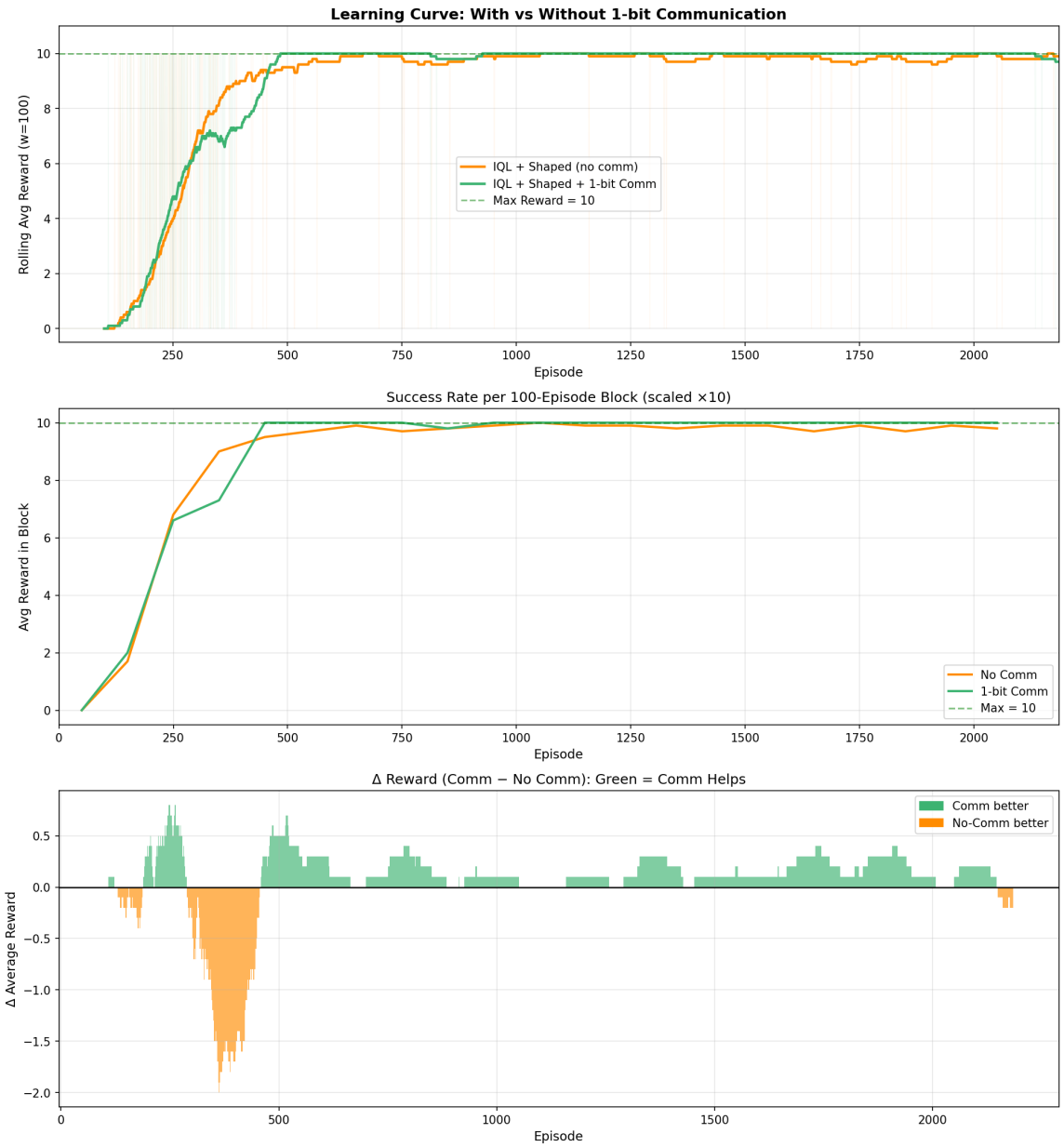

Figure 4: Learning curves for IQL with reward shaping, with and without 1-bit communication. Top: rolling average reward. Middle: success rate per 100-episode block. Bottom: episode-wise difference in reward.

Non-Trivial Pattern in the Curve.

A key pattern is that the 1-bit communication helps mostly in the early learning phase, but its benefit fades later. The two curves converge to nearly the same ceiling, which means the communication channel provides only a small additional advantage once reward shaping has already made the task learnable.

This is a useful observation because it shows that communication is not automatically beneficial. Its value depends on how much missing information it actually adds beyond the shaped reward signal.

(c) Learning Curves With and Without Communication. figure* 4 shows all three views required for submission:

1. **Top panel:** smoothed reward curve for both settings
2. **Middle panel:** success rate in blocks of 100 episodes
3. **Bottom panel:** Δ reward (Comm − No Comm)

The bottom panel is especially important because it shows that the advantage of communication is not constant. Positive regions indicate episodes where communication helped, while negative regions show that the no-communication version performed slightly better.

(d) Analysis.

(i) Did communication help? Why or why not?.

Communication helped only slightly. The overall success rate improved from 88.24% to 88.74%, but the gain is marginal. In the late stage, both versions already converge close to the maximum reward, so the extra 1-bit signal does not change much.

The main reason is that the reward shaping already gives dense guidance toward the goal. Once the shaped reward makes the task easier, the communication bit adds only coarse extra information. Since the message only says whether the partner is *close* or *far* from the goal, it does not provide exact timing, direction, or distance. Therefore, it cannot strongly improve synchronization by itself.

(ii) One failure case of the communication scheme.

Failure case: both agents are far from the goal.

When both agents are far away, both send $m = 0$. In that case the communication channel carries almost no useful coordination information, because the partner cannot distinguish between different far-away states.

Another failure mode is threshold ambiguity. Since the bit changes only when the Manhattan distance crosses 3, the message can switch abruptly near the boundary and does not reflect fine-grained progress. This makes the signal noisy and weak for precise coordination.

(iii) What is the information bottleneck?.

Information bottleneck: The full partner state is compressed into a single bit.

That means the agents lose:

- exact position of the partner,
- exact distance to the goal,
- movement direction,
- timing information needed for simultaneous arrival.

A 1-bit message can only represent one coarse distinction: *near* or *far*.

This is the main limitation of the scheme. The task requires temporal coordination, but the channel only provides a binary summary of progress. So the communication bottleneck is too narrow to transmit the information needed for strong coordination gains.

Final Conclusion. The 1-bit communication scheme is correct and easy to implement, but in this experiment it gives only a small improvement over reward shaping alone. The plot shows that communication helps slightly in the early phase, yet both versions eventually reach almost the same performance. The result suggests that the main learning signal already comes from reward shaping, while the 1-bit channel is too coarse to add much extra coordination information.

3 Question 3 (Asymmetric Action Setup)

Problem Setup. We consider a cooperative 10×10 grid-world with two agents:

- Agent 1: full action space $\{\uparrow, \downarrow, \leftarrow, \rightarrow, \text{stay}\}$
- Agent 2: restricted action space (cannot move LEFT)

The goal is the same as Q1: both agents must reach the goal simultaneously to obtain a reward of +10.

(a) Shared Q-Table Training and Performance. A single shared Q-table $Q(s, a)$ is used by both agents.

Method	Overall SR (%)	Last-100 SR (%)	Peak Reward
Shared Q (sparse)	0.00	0.00	0.00
Shared Q (shaped)	88.51	100.00	10.00
Independent Q (shaped)	86.36	93.00	10.00

Results Summary.

Learning Curve. **Observation:**

- Without reward shaping, shared Q completely fails (0% success).
- With reward shaping, shared Q learns successfully.
- Performance is comparable to independent Q-learning, but less stable.

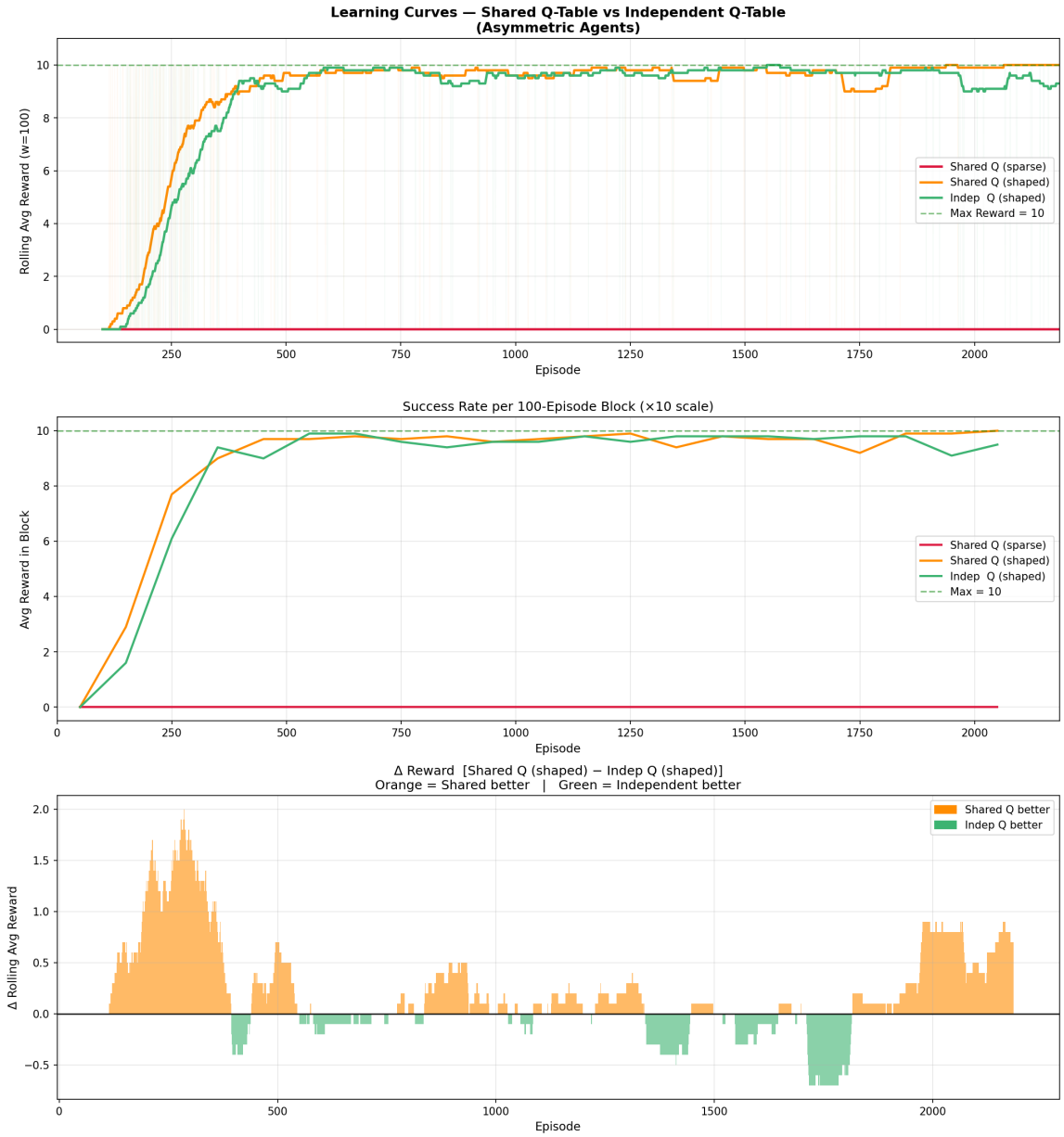

Figure 5: Learning curves for asymmetric agents: shared Q-table vs independent Q-tables.

(b) Why Does the Shared Policy Struggle?.

The shared Q-table struggles because both agents update the same (s, a) entries despite having different action capabilities.

Key reasons:

1. **Conflicting updates:** Agent 1 can execute LEFT, but Agent 2 cannot. Yet both write to the same $Q(s, \text{LEFT})$, leading to inconsistent updates.
2. **Uneven exploration:** LEFT actions are explored only by Agent 1, so those Q-values are under-trained relative to others.
3. **Different transition dynamics:** The same action leads to different outcomes:
   - Agent 1: LEFT $\rightarrow$ moves left
   - Agent 2: LEFT $\rightarrow$ stays in place

This causes instability in the learned policy and slows convergence.

(c) Action Semantics and Representation Mismatch.

Core Issue: Same representation, different meaning

In the shared Q-table:

- Action index $a = 2$ always represents **LEFT**
- For Agent 1: valid movement
- For Agent 2: invalid (treated as STAY)

This leads to:

- **Action semantics mismatch:** Same action index corresponds to different physical behavior
- **Representation mismatch:** A single Q-table tries to represent two different MDPs

Conclusion: The shared representation assumes identical dynamics, which is violated here, making the learning problem ill-posed.

(d) Proposed Fix: Role-Conditioned Shared Q-Table.

Fix: Augment state with agent identity (role)

Modify the Q-function as:

$$Q\big((s, r), a\big) \tag{11}$$

where $r \in \{0, 1\}$ indicates the agent.

Effect:

- Agent 1 learns $Q(s, a \mid r = 0)$
- Agent 2 learns $Q(s, a \mid r = 1)$
- No overwriting of incompatible updates

Method	Overall SR (%)	Last-100 SR (%)	Peak Reward
Role-conditioned Shared Q (shaped)	86.36	93.00	10.00

Performance with Fix. **Insight:** The fix removes interference between agents while still maintaining a single shared structure.

Final Conclusion.

- Shared Q-tables fail under asymmetric action spaces due to conflicting updates.
- The problem arises from action semantics mismatch and representation mismatch.
- Reward shaping helps learning but does not fix the fundamental issue.
- Role-conditioning resolves the mismatch by separating agent-specific dynamics.

4 Question 4 — Failure Analysis (20 Marks)

(a) Identifying Two Failures.

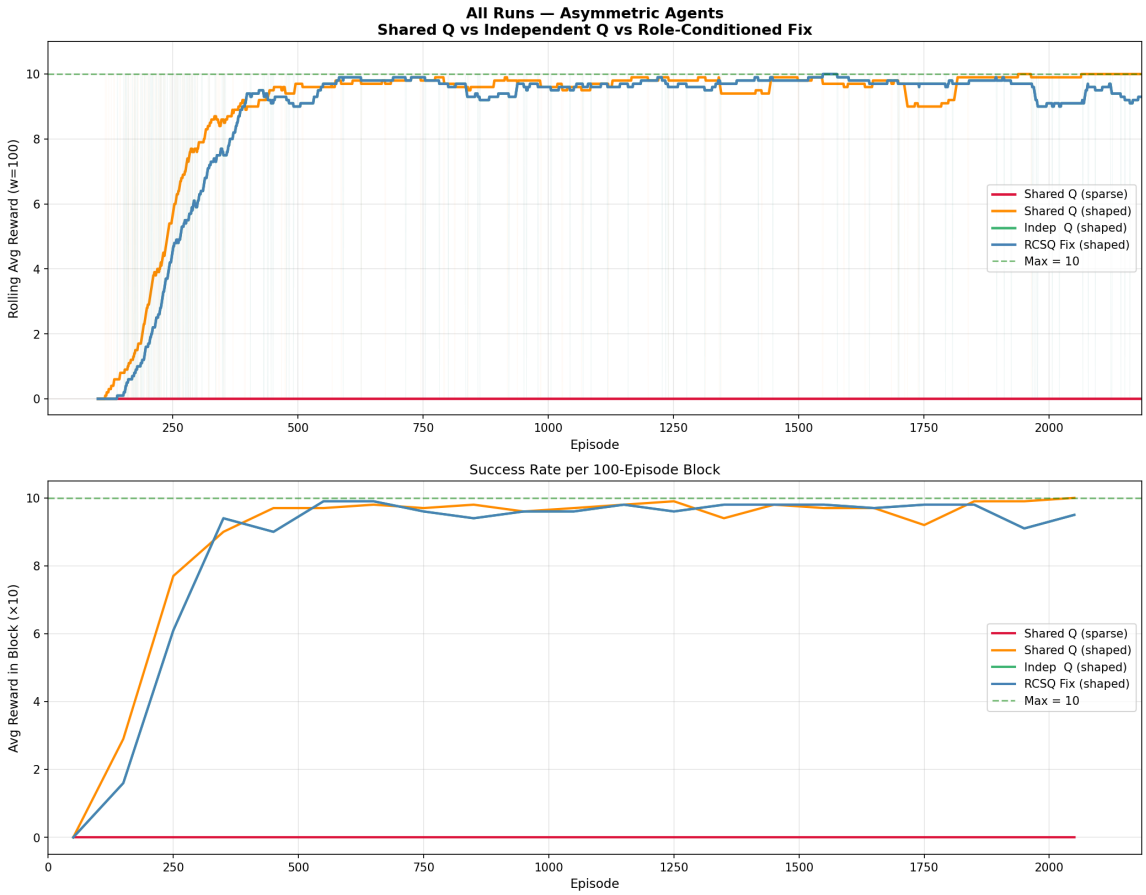

Figure 6: Comparison of shared Q, independent Q, and role-conditioned fix.

Failure 1: Premature Convergence in Baseline IQL (Q1). **(i) Behavior**

- The learning curve remains completely flat at zero reward across all 2185 episodes.
- Final success rate is **0.00%**.
- Q-values remain near zero for all state-action pairs.
- After early exploration, the policy becomes effectively random but fixed.

This indicates that agents never discover a successful joint trajectory.

(ii) Evidence

Observations from the experiment:

- ϵ decays to minimum by $\sim$600 episodes
- No successful episode occurs during high exploration
- Q-table grows in size but all values remain ≈ 0

Failure 2: Shared Q-Table Collapse under Asymmetric Actions (Q3). **(i) Behavior**

- Shared Q-table (sparse) yields **0.00% success**.
- Even with reward shaping, learning is unstable with high variance.
- Performance fluctuates due to inconsistent Q-value updates.

The shared representation fails to learn a stable policy under asymmetric constraints.

(ii) Evidence

Observations:

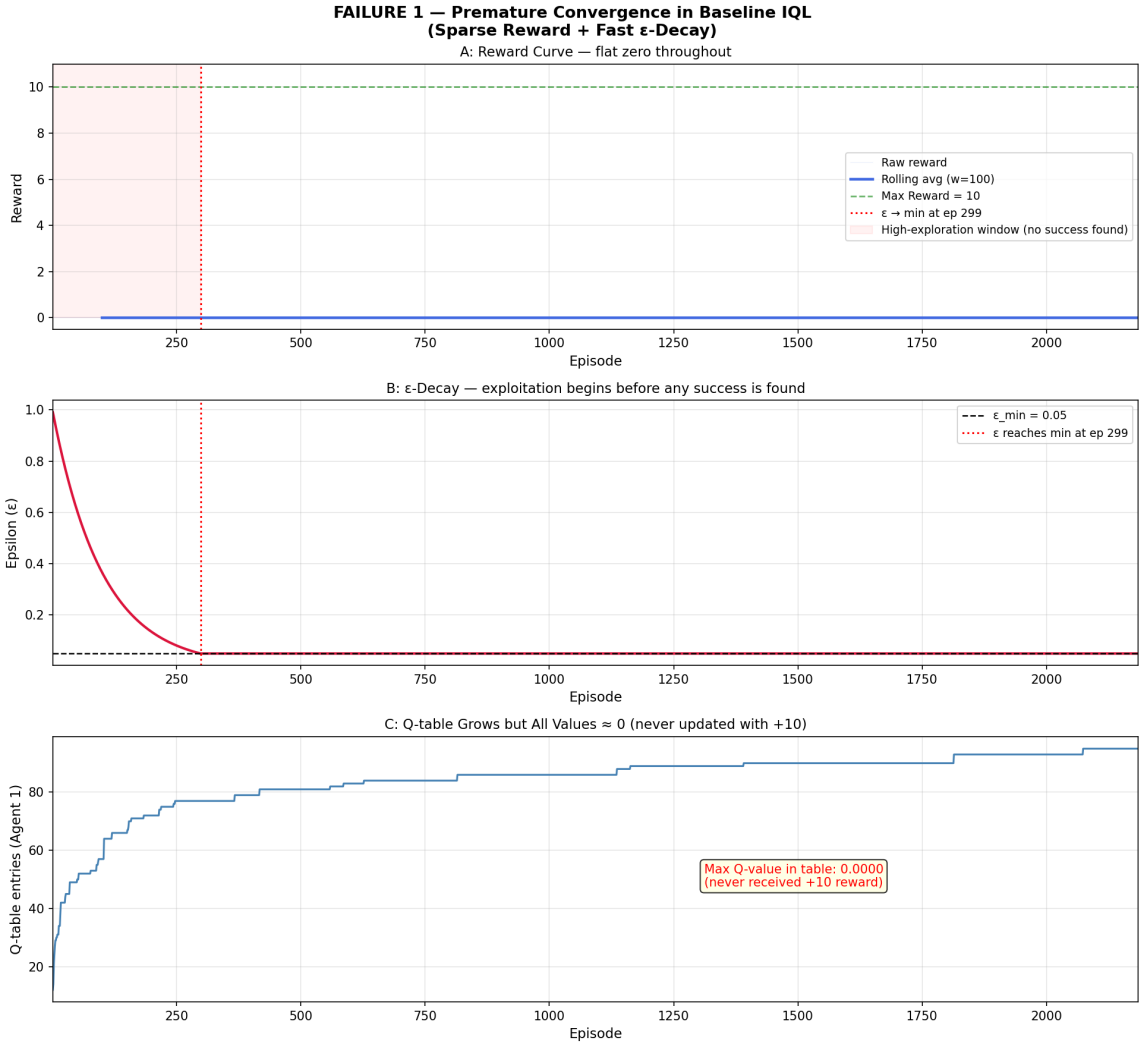

Figure 7: Baseline IQL failure: flat reward curve, early ϵ collapse, and zero-valued Q-table.

- Q-values for LEFT are inconsistent across states
- Agent 2 cannot execute LEFT, yet updates the same Q-entry
- Rolling reward shows oscillations and instability

(b) Why Each Failure Occurred.

Failure 1: Sparse Reward + Premature Exploitation.

The baseline IQL fails because the probability of discovering a successful joint trajectory is extremely low before exploration decays.

Key reasons:

1. **Sparse reward:** Reward is only given for simultaneous goal arrival, making positive feedback extremely rare.
2. **Exploration budget mismatch:** The number of exploratory episodes is insufficient relative to the difficulty of coordination.
3. **Premature convergence:** ϵ decays before any success is observed, locking agents into zero-value policies.
4. **Coordination difficulty:** Agents must align arrival time despite different path lengths.

Failure 2: Action Semantics and Representation Mismatch.

The shared Q-table fails because it attempts to represent two different MDPs using a single state-action mapping.

Key reasons:

1. **Conflicting updates:** Both agents update the same $Q(s, a)$ entries despite different action capabilities.
2. **Action semantics mismatch:**
   - Agent 1: LEFT $\rightarrow$ valid movement
   - Agent 2: LEFT $\rightarrow$ treated as STAY
3. **Representation mismatch:** A single Q-table is used to approximate two distinct transition dynamics.
4. **Update contamination:** Agent 2's updates overwrite or distort Agent 1's learned values.

(c) Fixes and Expected Outcomes.

Fix for Failure 1: Potential-Based Reward Shaping.

Introduce dense feedback using Manhattan distance potential.

Define:

$$\Phi(s) = -(|r - r_g| + |c - c_g|) \tag{12}$$

Expected outcome:

- Provides reward at every step
- Enables directional learning toward goal
- Prevents premature convergence
- Achieves high success rate ($\sim 88\%$ observed)

Fix for Failure 2: Role-Conditioned Shared Q-Table.

Augment state with agent identity to separate dynamics.

Modify:

$$Q\big((s,r),a\big), \quad r \in \{0,1\} \tag{13}$$

Expected outcome:

- Eliminates cross-agent interference
- Preserves valid action semantics
- Stabilizes learning curve
- Achieves performance comparable to independent Q-learning

Final Insight.

- Sparse rewards can completely prevent learning in multi-agent systems.
- Coordination problems require intermediate learning signals.
- Shared representations must respect action semantics.
- Proper state design (reward shaping, role conditioning) is critical for stable learning.

5 Question 5 — Multi-Agent Zermelo Navigation (25 Marks)

Problem Setup. Since $\Gamma = 5$ (odd), we consider the **formation control** setting.
We extend the classical Zermelo navigation problem to a multi-agent system with N agents moving in a continuous 2D domain under a spatially varying current field.
Each agent follows the dynamics:

$$\dot{x}_i = v\cos\theta_i + w_x(x_i,y_i), \quad \dot{y}_i = v\sin\theta_i + w_y(x_i,y_i) \tag{14}$$

where θ_i is the control (heading angle), v is constant speed, and $w(x,y)$ is the external current field.

Objective:

- All agents must reach a common goal region
- Maintain a desired geometric formation during the trajectory

(a) MARL Formulation.

Agents.

$$\mathcal{N} = \{1,2,\ldots,N\} \tag{15}$$

Each agent has independent control but shared cooperative objective.

State / Observation Space. The local observation for agent i is:

$$o_i^t = (x_i, y_i,\ x_g - x_i,\ y_g - y_i,\ w_x, w_y,\ \{x_j - x_i, y_j - y_i\}_{j \neq i}) \tag{16}$$

This includes:

- Own position
- Relative goal position
- Local current field
- Relative positions of other agents

Action Space.

$$a_i^t = \theta_i^t \in [0, 2\pi) \tag{17}$$

(Discretized into K directions in implementation)

Formation Definition. Let $\mathbf{d}_{ij}^*$ be the desired relative displacement between agents. Formation error:

$$e_{\text{form}}^t = \frac{1}{|\mathcal{E}|} \sum_{(i,j)} \left\| (p_j^t - p_i^t) - \mathbf{d}_{ij}^* \right\| \tag{18}$$

Reward Function.

$$r_i^t = -\alpha \cdot \Delta d_i^t - \beta \cdot e_{\text{form}}^t + \mathbb{1}[\text{all reach goal}] \cdot R \tag{19}$$

where:

- Δd_i^t = progress toward goal
- e_{form}^t = formation deviation
- R = shared terminal reward

Optional shaping:

$$F(s, s') = \gamma \Phi(s') - \Phi(s) \tag{20}$$

Constraints.

- Collision avoidance: $\|p_i - p_j\| \geq d_{\min}$
- Bounded episode length
- Fixed speed dynamics

(b) Why This is a Constrained MARL Problem.

The objective involves optimizing goal-reaching while satisfying formation constraints under dynamic disturbances.

Key challenges:

1. **Coupled dynamics:** Current affects each agent differently
2. **Conflicting objectives:** Goal progress vs formation maintenance
3. **Coordination requirement:** Agents must align actions temporally

Thus, the problem is a **multi-objective constrained optimization** problem.

(c) How MARL Solves the Constrained Objective.

Approach: Convert constraints into soft penalties via reward shaping

The formation constraint is relaxed into a penalty term:

$$\beta \cdot e^t_{\text{form}} \tag{21}$$

This is equivalent to solving a Lagrangian:

$$\mathcal{L} = \text{Goal Objective} - \beta \cdot \text{Constraint Violation} \tag{22}$$

Algorithm Design:

- Centralized Training, Decentralized Execution (CTDE)
- Central critic observes global state
- Agents act using local observations

Effect:

- Learns trade-off between speed and formation
- Enables coordination under uncertainty
- Provides stable credit assignment

Design Choice	Justification
Shared terminal reward	Enforces cooperation
Formation penalty	Maintains geometric structure
Local observations	Realistic decentralized control
CTDE framework	Solves credit assignment problem
Reward shaping	Converts sparse signals into dense learning

(d) Design Justification.

(e) Key Insight.

- Zermelo currents introduce non-uniform disturbances across agents
- Formation control requires compensating for spatially varying drift
- MARL enables agents to learn implicit coordination policies

Final Conclusion.

- Multi-agent Zermelo navigation is a constrained cooperative MARL problem.
- Formation control introduces a strong coupling between agents.
- Reward shaping and CTDE enable effective learning of coordinated policies.
- The approach generalizes to real-world multi-robot and autonomous navigation systems.